Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 11: Waveguides 3

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Cutoff frequency

$$\begin{cases} \nabla_t^2 \psi + \gamma^2 \psi = 0 & \psi = E_z \text{ or } B_z \\ \psi = 0 \quad \text{or} \quad \dfrac{\partial \psi}{\partial n} = 0 & \text{on } S \end{cases} \qquad \gamma^2 = \mu\varepsilon\omega^2 - k^2$$

γ_λ^2 eigenvalue, ψ_λ eigenfunction.

$\lambda = 1, 2, 3, \ldots$

The solutions are called modes of the waveguide.

Cutoff frequency

r_λ is solved from the 2D Laplace

$$\gamma^2 = \varepsilon\mu\omega^2 - k^2 \qquad k^2 = \varepsilon\mu\omega^2 - \gamma^2$$

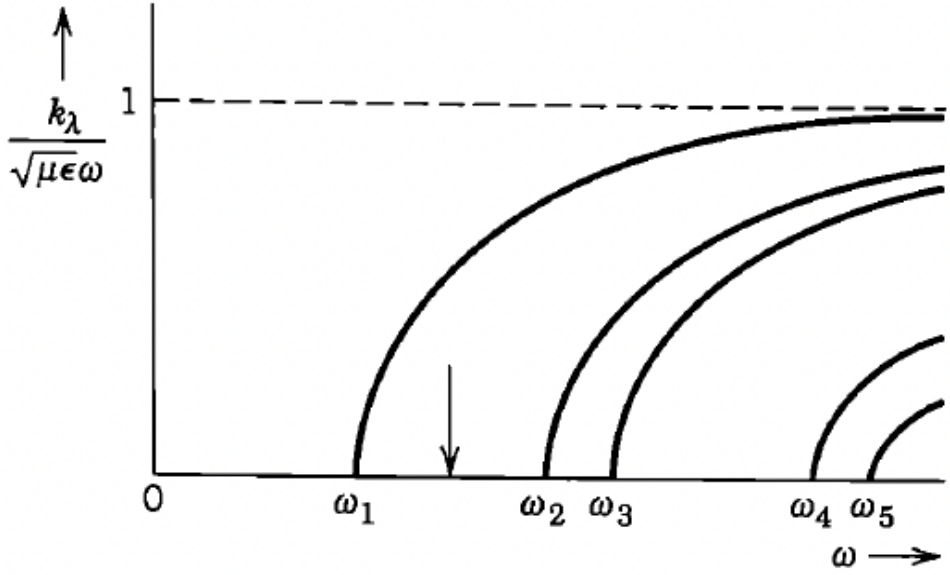
$$\vec{E}, \vec{B} \propto e^{ikz - i\omega t}$$

k must be real for the wave to propagate

$\Rightarrow k^2 > 0$ introduce $\omega_\lambda = \dfrac{r_\lambda}{\sqrt{\mu\varepsilon}}$ cutoff frequency

$$k^2 = \varepsilon\mu\omega^2 - \varepsilon\mu\omega_\lambda^2 \Rightarrow k = \sqrt{\mu\varepsilon}\sqrt{\omega^2 - \omega_\lambda^2}$$

Cutoff frequency



vp and vg at cutoff frequency

$$v_p = \frac{\omega}{k}$$

$$= \frac{\omega}{\sqrt{\mu\varepsilon}\,\sqrt{\omega^2-\omega_\lambda^2}}$$

$$= \frac{c}{n}\quad\frac{\omega}{\sqrt{\omega^2-\omega_\lambda^2}} > \frac{c}{n}$$

$$v_g = \frac{d\omega}{dk}$$

$$\omega = \sqrt{\omega_\lambda^2 + \frac{k^2}{\mu\varepsilon}}$$

$$v_g = \frac{d\omega}{dk} = \frac{k/\mu\varepsilon}{\sqrt{\omega_\lambda^2 + \frac{k^2}{\mu\varepsilon}}}$$

$$= \frac{k}{\mu\varepsilon\omega} = \frac{1}{v_p\mu\varepsilon}$$

$$\Rightarrow\quad v_g \cdot v_p = \frac{1}{\mu\varepsilon}$$

Example: TE in Rectangular Waveguide

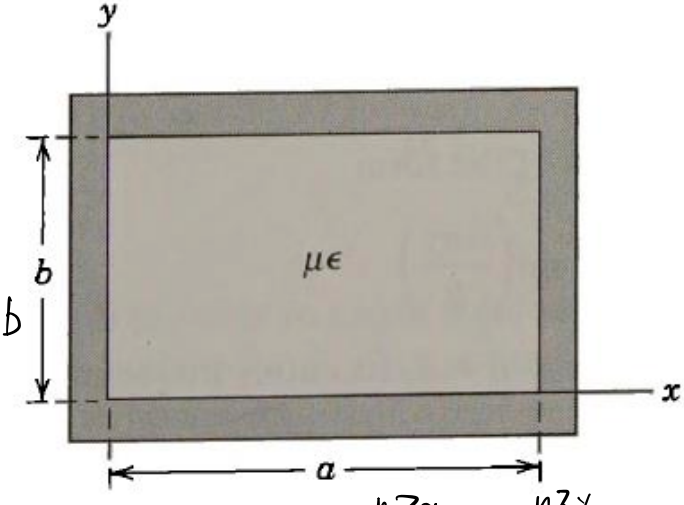
$E_z = 0$, $H_z \neq 0$

- 2D wave eq

$$\begin{cases} \left(\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + \gamma^2\right) H_z = 0 \\ \dfrac{\partial H_z}{\partial n} = 0 \quad \text{on} \quad x = 0, a, \; y = 0, b \end{cases}$$

$$\frac{\partial H_z}{\partial x} = 0 \quad \text{on} \quad y = 0, b$$

$$\frac{\partial H_z}{\partial y} = 0 \quad \text{on} \quad x = 0, a$$



$$\psi_{mn}(x,y) = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$$

$$\gamma_{mn}^2 = \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)$$

Rectangular Waveguide

- Cutoff freq

$$\omega_{mn} = \frac{\pi}{\sqrt{\mu\epsilon}} \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right)^{1/2}$$

- Lowest $a > b$ $m = 1,\ n = 0$

$$\omega_{10} = \frac{\pi}{\sqrt{\mu\epsilon}} \frac{1}{a} = \frac{\pi}{a} \frac{c}{\eta}$$

Rectangular Waveguide

- Lowest solution

$$H_z = H_0 \cos\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t}$$

$$\vec{E}_t = \frac{i\omega}{\gamma^2}\left[-\omega \hat{z} \times \nabla_t H_z\right] \quad \hat{y}$$

$$\vec{H}_t = \frac{i}{\gamma^2}\left[ik \nabla_t H_z\right] \quad \propto \hat{x}$$

$$\nabla_t H_z = \frac{\partial H_z}{\partial x}\hat{x} = -H_0 \frac{\pi}{a} \sin\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t} \hat{x}$$

Rectangular Waveguide

- Lowest solution

$$
\begin{cases}
H_z = H_0 \cos\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t} \\
H_x = -\frac{ika}{\pi} H_0 \sin\frac{\pi x}{a} e^{ikz - i\omega t} \\
H_y = 0 \\
\bar{b}_x = \bar{b}_z = 0 \\
\bar{b}_y = i\frac{\omega a \mu}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{ikz - i\omega t}
\end{cases}
$$

TM in Rectangular Waveguide?

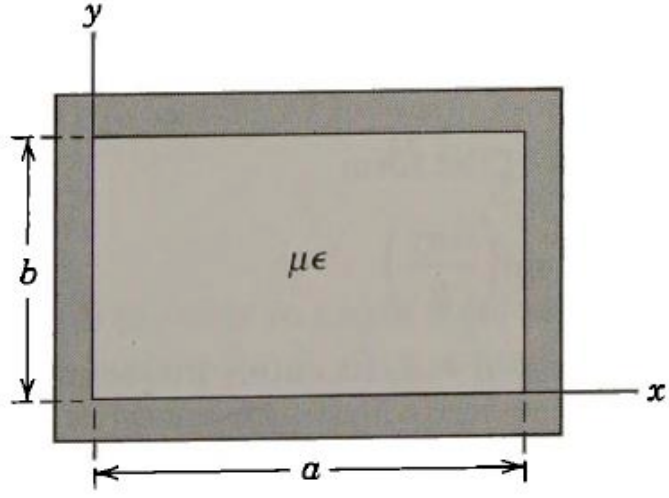
$H_z = 0$

- 2D wave eq

$$\begin{cases} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \gamma^2\right) \bar{E}_z = 0 \\ \bar{E}_z = 0 \quad \text{on} \quad x = 0, a \cdot y = 0, b \end{cases}$$

$$\psi_{mn} = \bar{E}_0 \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\gamma_{mn}^2 = \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)$$



$$\omega_{mn} = \frac{\pi}{\sqrt{\varepsilon\mu}} \left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)^{1/2}$$

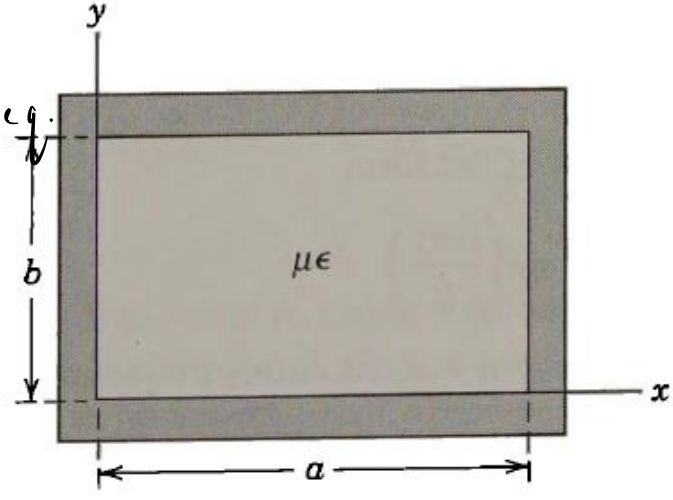
TM in Rectangular Waveguide?

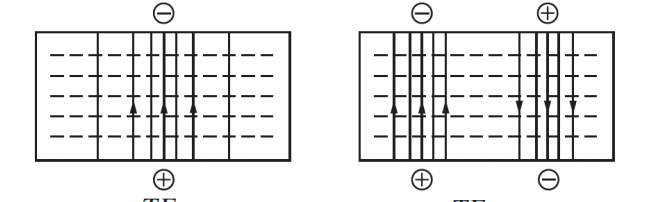
- 2D wave eq

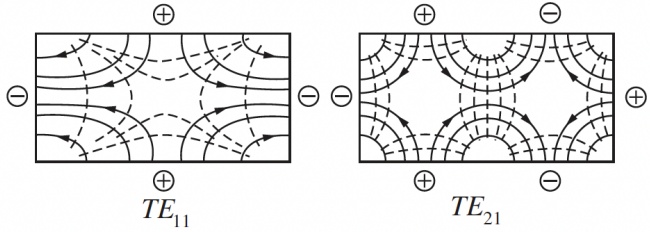
$m = n = 1 \Rightarrow$ Lowest cutoff freq.

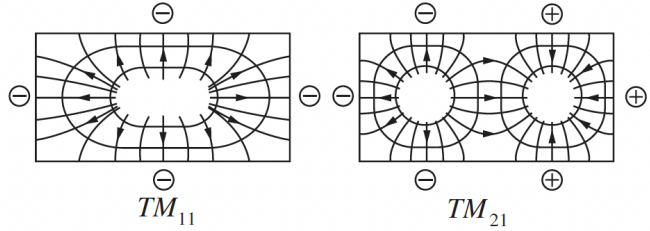
$$\omega_{11} = \frac{\pi}{\sqrt{\mu\epsilon}} \left(\frac{1}{a^2} + \frac{1}{b^2} \right)^{1/2}$$

$$\omega_{TE,\,Lowest} < \omega_{TM,\,Lowest}$$

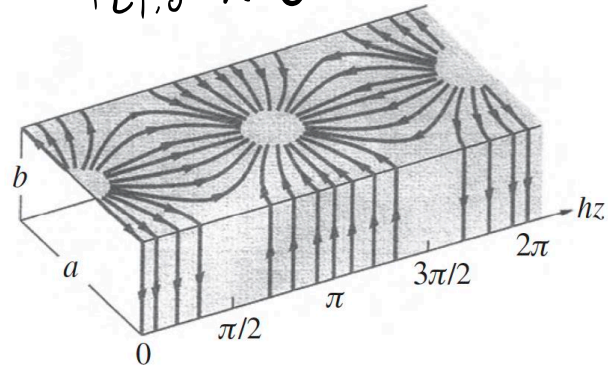








$TE_{1,0}$ mode



Energy Flow in Waveguides

- Poynting vector

$$\langle \vec{S} \rangle = \frac{1}{2} \vec{E} \times \vec{H}^*$$

TM: $\langle \vec{S} \rangle = \frac{1}{2} (\vec{E}_z + \vec{E}_t) \times \vec{H}_t^*$

TE: $\langle \vec{S} \rangle = \frac{1}{2} \vec{E}_t \times (\vec{H}_t + \vec{H}_z)^*$

$\psi = E_z$ or H_z $\quad \vec{H}_t(\psi)$, $\vec{E}_t(\psi)$

Energy Flow in Waveguides

- Poynting vector

TM: $\langle \vec{S} \rangle = \frac{1}{2} (\vec{E}_t + \vec{E}_z) \times \vec{H}_t^*$

$= \frac{1}{2} \left(\frac{i}{\gamma^2} k \nabla_t \psi + \psi \hat{z} \right) \times \frac{-i}{\gamma^2} \left[\varepsilon \omega \hat{z} \times \nabla_t \psi^* \right]$

$= \frac{k \varepsilon \omega}{2\gamma^4} \nabla_t \psi \times (\hat{z} \times \nabla_t \psi^*) + \frac{\varepsilon \omega}{2} \frac{i}{\gamma^2} \psi \nabla_t \psi^*$

$= \frac{\omega k}{2\gamma^4} \varepsilon \left[\hat{z} |\nabla_t \psi|^2 + i \frac{\gamma^2}{k} \psi \nabla_t \psi^* \right]$

Energy Flow in Waveguides

HW TE:

$$\langle \vec{S} \rangle = \frac{\omega k}{2\gamma^4} u \left[\bar{z} \left| \nabla_t \psi \right|^2 - i \frac{\gamma^2}{k} \psi^* \nabla_t \psi \right]$$

$$\langle S_z \rangle = \langle \vec{S} \rangle \cdot \vec{z} = \frac{\omega k}{2\gamma^4} \begin{Bmatrix} \varepsilon \\ u \end{Bmatrix} \left| \nabla_t \psi \right|^2 \quad \begin{matrix} TM \\ TE \end{matrix}$$

$$\langle P \rangle = \int_A \langle S_z \rangle \, da$$

Energy Flow in Waveguides

$$\langle P \rangle = \int_A \langle S_z \rangle \, da$$

$$= \frac{\omega k}{2\gamma^4} \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \int_A (\nabla_t \psi)^* \cdot (\nabla_t \psi) \, da$$

Green's 1st identity

$$= \frac{\omega k}{2\gamma^4} \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \left[\oint_A \psi^* \frac{\partial \psi}{\partial n} \, dL - \int_A \psi^* \nabla_t^2 \psi \, da \right]$$

$\psi = 0$ or $\frac{\partial \psi}{\partial n} = 0$ on S

Energy Flow in Waveguides

$$= \frac{\omega k}{2\gamma^4} \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \left[\oint_A \psi^* \frac{\partial \psi}{\partial n} dL - \int_A \psi^* \nabla_t^2 \psi \, da \right]$$

$\psi = 0$ or $\frac{\partial \psi}{\partial n} = 0$ on S $\quad \Rightarrow \oint_A = 0$

$$\nabla_t^2 \psi + \gamma^2 \psi = 0$$

$$\langle P \rangle = \frac{\omega k}{2\gamma^2} \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \int \psi^* \psi \, da \qquad k = \sqrt{\mu\varepsilon}\sqrt{\omega^2 - \omega_\lambda^2}$$

$$\gamma^2 = \mu\varepsilon\omega_\lambda^2$$

$$= \frac{1}{2\sqrt{\mu\varepsilon}} \left(\frac{\omega}{\omega_\lambda}\right)^2 \left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)^{1/2} \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \int \psi^* \psi \, da$$